

Roll No.

TPH-201

**B. TECH. (SECOND SEMESTER)
END SEMESTER EXAMINATION, June, 2024**

ENGINEERING PHYSICS

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.
1. (a) Explain interference and write the essential conditions required to obtain sustained interference. In Newton's ring experiment, deduce the expression for diameter of dark rings. (CO1)
(b) In Fraunhofer diffraction due to N-slits derive the expression for resultant intensity and also write the ratio of intensities of principal maxima to secondary maxima. (CO1)
(c) (i) Derive the expression for resolving power of grating.
(ii) Deduce the expression for fringe width in a wedge-shaped film. (CO1)
 2. (a) Derive the necessary relation between Einstein coefficients for absorption and stimulated emission. Explain how the He-Ne laser is superior to the Ruby laser. (CO2)
(b) What is plane, elliptically, and circularly polarized light ? Discuss the method for their production using necessary mathematical expressions. (CO2)

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- (c) What is specific rotation ? Discuss the construction and working of Laurent's Half shade polarimeter. Two polaroids are adjusted to obtain maximum light; through what angle should one polaroid be rotated to reduce the intensity to $1/4$ th ? (CO2)
3. (a) Deduce Lorentz transformation equation of space and time coordinates. Find the kinetic energy of mesons having a mass of 2.4×10^{-28} kg traveling with the relativistic speed of $0.8 c$. (CO3)
- (b) Derive the expression for time dilation; also show its experimental verification. Find the relative speed of two photons moving in opposite directions. (CO3)
- (c) Derive Einstein's mass-energy relation. Calculate the percentage of contraction in the length of a rod moving with a relativistic speed of $0.8 c$ in a direction at an angle of 60 degrees with its length. (CO3)
4. (a) Show that the solution of Schrodinger's equation for a particle in an infinite potential well leads to the concept of quantization of energy. Also depict graphically the probability of locating the particle in the first three energy levels $n = 1, 2, 3$. (CO4 & CO5)
- (b) An electron is confined in a potential well of width 10 nm. Calculate the minimum uncertainty in its velocity. What do you mean by normalized wave function; also write the necessary condition for physically acceptable wave function. (Given : mass of an electron = 9.1×10^{-31} kg and $h = 6.63 \times 10^{-34}$ J-sec). (CO4 & CO5)
- (c) Explain the following : Quantum bits (qubit)superposition, qubit entanglement, and nano qubit implementation. (CO4 & CO5)
5. (a) What is the significance of critical temperature and critical magnetic field for superconductors ? What do you mean by perfect diamagnetism of superconductors ? (CO5 & CO6)
- (b) Derive all four Maxwell's equations and write their physical significance. Also, write the integral form of these equations in free space. (CO5 & CO6)
- (c) Explain the role of quantum confinement and the different structures formed by it at the nanoscale and also give the details of the ball milling method. (CO5 & CO6)

End Semester Examination
June 2024

Name of the Program: B Tech
Name of the Course: Engineering Chemistry

Semester: 2nd
Course Code: TCH201

Time: 3 Hr

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1 (20 marks)		CO 1
a)	(i) What are the wavelength ranges for UV, IR and Visible regions? Distinguish between atomic spectra and molecular spectra. (ii) Write applications of UV and IR spectroscopy	
b)	(i) Differentiate between BMO and ABMO (ii) What do you understand by extrinsic and intrinsic semiconductors? Explain conduction in p-type and n-type semiconductors.	
c)	Draw the Molecular orbital diagram of CO and NO and calculate the bond order. Also, mention which molecular orbital is the HOMO and which is the LUMO?	
Q2 (20 marks)		CO 2
a)	Distinguish between demineralized water and soft water. How is water softened by the Ion exchange method? Explain in detail with a neat diagram, structure of resin, chemical equations and regeneration of ion exchange resin.	
b)	(i) A water sample contains the following impurities. Ca(HCO ₃) ₂ – 32.4 ppm, Mg(HCO ₃) ₂ – 14.6 ppm, CaSO ₄ – 13.6 ppm, MgSO ₄ – 24.0 ppm, NaCl – 50.0 ppm. Calculate temporary and permanent hardness in water. Also report hardness in °Cl and °Fr. (ii) A zeolite softener was 90% exhausted, when 20,000 L of hard water was passed through it. The softener then required 300 L of NaCl solution having strength 50 g/L for regeneration. What is the hardness of water?	
c)	What are the requirements of boiler feed water? Write notes on i) Caustic embrittlement ii) Priming and foaming iii) Calgon conditioning iv) Temporary and permanent hardness	
Q3 (20 marks)		CO 3
a)	What do you mean by conducting polymers? List out the applications of conducting polymers.	

b)	(ii) Write note on biodegradable polymers by giving any two examples. (ii) Differentiate between thermoplastic and thermosetting plastics. Describe the preparation, properties and uses of Bakelite and PVC	
c)	Explain the following mechanisms: (i) Free radical addition polymerization. (ii) Cationic addition polymerization.	
Q4 (20 marks)		
a)	(i) Mention the chemical composition and properties of CNG. How is CNG superior to LPG? (ii) compare the merits and demerits of solid liquid and gaseous fuels.	CO4
b)	What is the difference between GCV and NCV. The following data is obtained in a bomb calorimeter experiment: Mass of crucible = 3.649 gm Mass of crucible with fuel = 4.678 gm Water equivalent mass of calorimeter = 0.57 kg Mass of water = 2.2 kg Rise in temperature = 2.5 °C Cooling correction = 0.047 °C Fuse wire correction = 11 cal Cotton thread correction = 10 cal Acid Correction = 12 cal Calculate GCV of fuel sample. Determine NCV If the fuel contains 6% H.	
c)	List the raw material which can be utilized for biogas manufacture. Describe manufacture of biogas with neat diagram and give its chemical composition.	
Q5 (20 marks)		
a)	Explain the construction and working of Galvanic cell with a neat diagram. Derive Nernst equation for EMF of a cell. Mention its applications.	CO5
b)	(i) Define the term standard electrode potential. Can we use a nickel spatula to stir a solution of CuSO ₄ ? ($E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +0.34\text{V}$, $E^\circ_{\text{Ni}^{2+}/\text{Ni}} = +0.025\text{V}$). Give reason for your answer. (ii) What is a concentration cell? How is this different from Galvanic cell?	
c)	(i) Following reactions occur at cathode during the electrolysis of aqueous sodium chloride solution: $\text{Na}^+(\text{aq}) + e^- \longrightarrow \text{Na}(\text{s}) \quad E^\circ = -2.71 \text{ V}$ $\text{H}^+(\text{aq}) + e^- \longrightarrow 1/2\text{H}_2(\text{g}) \quad E^\circ = 0.00 \text{ V}$ On the basis of their standard reduction electrode potential (E°) values, which reaction is feasible at the cathode and why? (ii) Calculate E° and G° for the following reaction at 25 °C: $\text{A}^{2+}(\text{aq}) + \text{B}^+(\text{aq}) \longrightarrow \text{A}^{5+}(\text{aq}) + \text{B}(\text{s})$ Given $K_c = 10^{10}$ Moles/Liter	

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EVEN END SEMESTER EXAMINATION - 2024

Name of the Course: B. Tech Bio Tech
Name of the Paper Engineering Mathematics

Semester: II
Paper Code: TMA 202

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub-questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1	(20 marks)	CO1
(a)	Find $A^2 - 5A + 6I$, if $A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$.	
(b)	Find the Eigen values and Eigen vectors of the given matrix: $\begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$	
(c)	Find the inverse of the given matrix : $A = \begin{bmatrix} 4 & -2 & 1 \\ 7 & 3 & 3 \\ 2 & 0 & 1 \end{bmatrix}$	
Q2	(20 marks)	CO2
(a)	Evaluate: $\int x \log x \, dx$ and $\int x^7 e^{6x} dx$	

(b)	Solve the following system of Linear equation: $x + 2y + z = 7; x + 3z = 11; 2x - 3y = 1$	
(c)	Find the derivative of the function $y = x^x + x^a + a^x + a^a$	
Q3	(20 marks)	
(a)	Solve: $(x + 1) \frac{dy}{dx} - y = e^x(x + 1)^2$.	
(b)	Define the linear differential equation and solve $\frac{dy}{dx} = e^{x-y} + x^3 e^{-y}$.	CO3
(c)	Solve $(x^2 - y^2) dx + 2xy dy = 0$	
Q4	(20 marks)	
(a)	Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using Simpson's one third rule and Simpson's three eight rule.	CO4
(b)	Find a real root of the equation $x^4 - x = 10$ by Newton -Raphson method.	
(c)	Find a real root of the equation $x^3 - 2x - 5 = 0$ by the false position method correct to three decimal places.	
Q5	(20 marks)	
(a)	A factory production line is manufacturing bolts using three machines, A, B and C. Of the total output, machine A is responsible for 25%, machine B for 35% and machine C for the rest. It is known from previous experience with the machines that 5% of the output from machine A is defective, 4% from machine B and 2% from machine C. A bolt is chosen at random from the production line and found to be defective. What is the probability that it came from? (a) Machine A (b) machine B (c) machine C?	CO5
(b)	A candidate is selected for interview of management trainees for 3 companies. For the first company there are 12 candidates, for the second there are 15 candidates and for the third there are 10 candidates. What are the chances of his getting job at least at one of the company?	
(c)	In a test, an examinee either guesses or copies or knows the answer to a multiple-choice question with four choices. The probability that he makes a guess is $1/3$ and the probability that he copies the answer is $1/6$. The probability that his answer is correct, given that he copied it is $1/8$. Find the probability that he knows the answer to the question, given that he correctly answered it.	

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EVEN END SEMESTER EXAMINATION-2024

Name of the Course: B.Tech

Semester: II

Name of the Paper: Engineering Mathematics-II

Paper Code: TMA 201

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Each main question carries 20 marks.
- (iv) Each sub-question carries 10 marks.

Q1	(20 marks)	
(a)	Test the given equation is exact or not. Also find the solution. $2ydx + (2x\log x - xy)dy = 0$	CO 1
(b)	Solve $(D^2 - 4D + 4)y = x^3e^{2x}$; $D = \frac{d}{dx}$.	
(c)	Solve $(D^2 + a^2)y = \operatorname{cosec} ax$ by method of variation of parameters.	
Q2	(20 marks)	
(a)	Using Convolution theorem, find inverse Laplace transform of the function $f(s) = \left\{ \frac{s}{(s^2+4)^2} \right\}$.	CO 2
(b)	Evaluate $\int_0^\infty \frac{e^{-t}\sin t}{t} dt$.	
(c)	Solve the given equation by Laplace transform $\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 0$; where $y(0) = 1, y'(0) = 2, y''(0) = 2$.	
Q3	(20 marks)	
(a)	Solve the PDE : $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 2\frac{\partial^2 z}{\partial y^2} = (y-1)e^x$	CO 3
(b)	Solve the PDE by Lagrange method $yq - xp = z$.	
(c)	Solve the non-linear PDE by Charpit's method $z = px + qy + p^2 + q^2$	
Q4	(20 marks)	
(a)	Solve $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6\frac{\partial^2 z}{\partial y^2} = x + y$.	
(b)	Classify the following PDE's: (i) $2\frac{\partial^2 u}{\partial x^2} + 4\frac{\partial^2 u}{\partial x \partial t} + 3\frac{\partial^2 u}{\partial t^2} = 0$. (ii) $9\frac{\partial^2 u}{\partial x^2} - 6\frac{\partial^2 u}{\partial x \partial t} + \frac{\partial^2 u}{\partial t^2} = 0$. (iii) $x^2\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = u$. (iv) $x\frac{\partial^2 u}{\partial t^2} + t\frac{\partial^2 u}{\partial x \partial t} + \frac{\partial^2 u}{\partial t^2} = 0$.	CO 4
(c)	Solve: $\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x} - 2u$, $u(x, 0) = 10e^{-x} - 6e^{-4x}$	

Q5	(20 marks)	CO 5
(a)	Find the Fourier series representing $f(x) = x \cos x, -\pi < x < \pi$.	
(b)	Expand $\pi x - x^2$ in half range sine series in the interval $(0, \pi)$ upto first three terms.	
(c)	Find the Fourier sine transformation of $\frac{e^{-ax}}{x}$	

End Semester Examination June 2024

Name of the course: B. Tech

Semester: II

Name of the Paper: Basic Electronics Engineering

Paper Code: **TEC 201**

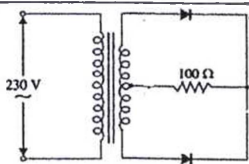
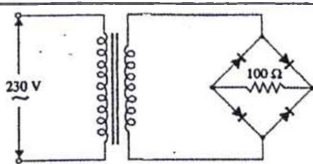
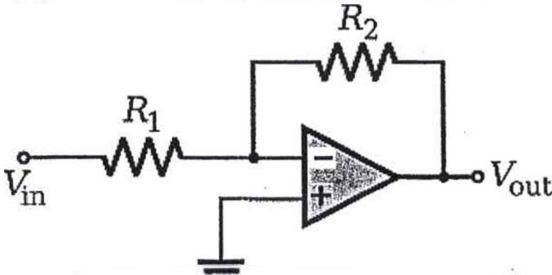
Time: 3Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory
- (ii) Answer any two **questions** among a, b, & c **in each main question**.
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	List all the basic and universal gates with their symbol and truth table. Realize EX-NOR gate using NAND gates only.	CO1
(b)	Obtain the minimal SOP expression, with the help of k-map, for the following function and implement the same using NOR gate only: $F(A, B, C, D) = \sum m(0,1,3,4,5,7,12,13,15) + \sum d(8,9)$	
(c)	Simplify the given Boolean expression using Boolean rules: i) $F = ABD + CD + ACD + ABC + ABCD$ ii) $F = (A+C)(AD + AC + C)$ Convert the base of following numbers: iii) $(634512)_7 = (?)_{10}$ iv) $(943784)_{10} = (?)_4$	
Q2	(20 marks)	
(a)	Explain the forward and reverse bias condition of a PN junction diode with V-I characteristics.	CO2
(b)	(i) Distinguish between Avalanche breakdown and Zener breakdown. (ii) A Si diode has reverse saturation current of $2.8 \mu A$ at $300K$. Find forward voltage to achieve a forward current of $12 mA$. ($V_T = 26 mV$)	
(c)	A sample of pure Ge has intrinsic charge carrier density of $2.5 \times 10^{19}/m^3$. If it is doped with donor impurity of 1 atom in every 10^6 Ge atom. i) Determine majority and minority charge carriers if, $\mu_n = 0.35 m^2/V\text{-sec}$ and $\mu_p = 0.14 m^2/V\text{-sec}$; Ge atom density is $4.4 \times 10^{28}/m^3$. ii) Conductivity and resistivity of doped semiconductor. iii) If the length of semiconductor is $5.5 mm$ and cross section area is $20 \times 10^{-12} m^2$, find resistance.	
Q3	(20 marks)	
(a)	Fig. (i) and Fig. (ii) show the centre-tap and bridge type circuits having the same load resistance (100Ω) and transformer turn ratio ($10:1$). The primary of each is connected to $230V$, $50 Hz$ supply. Find peak, avg. and rms current in each case. Assume the diodes to be ideal with forward resistance of 25Ω .	CO3

	 (i)  (ii)	
(b)	Explain the working of half wave rectifier with appropriate circuit as well as input/output waveforms. Derive an expression for its efficiency as well.	
(c)	Derive an expression for dc and rms voltage for a full wave rectifier.	
Q4	(20 marks)	
(a)	With help of appropriate circuit diagrams, explain the input as well as output characteristics of Common Base Configuration of PNP or NPN transistor.	CO4
(b)	With help of appropriate circuit diagrams, explain the input as well as output characteristics of Common Emitter Configuration of PNP or NPN transistor.	
(c)	Derive the relationship between: (i) α , β and γ (ii) I_{CEO} and I_{CBO} .	
Q5	(20 marks)	
(a)	i) What is summing amplifier? Derive an expression for its output voltage. ii) Determine the value of feedback resistor R_2, such that the close loop gain is -5 and $R_1 = 15 \text{ k}\Omega$, in the circuit given below. 	CO5/CO6
(b)	Explain the inverting and non-inverting amplifier. Also, derive an expression for their gain as well.	
(c)	Derive an expression for output voltage of difference amplifier.	



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EVEN END SEMESTER EXAMINATION 2024

Name of the Course: B. Tech Biotech

Semester: II

Name of the Paper: Advanced Organic Chemistry

Paper Code: TCH-202

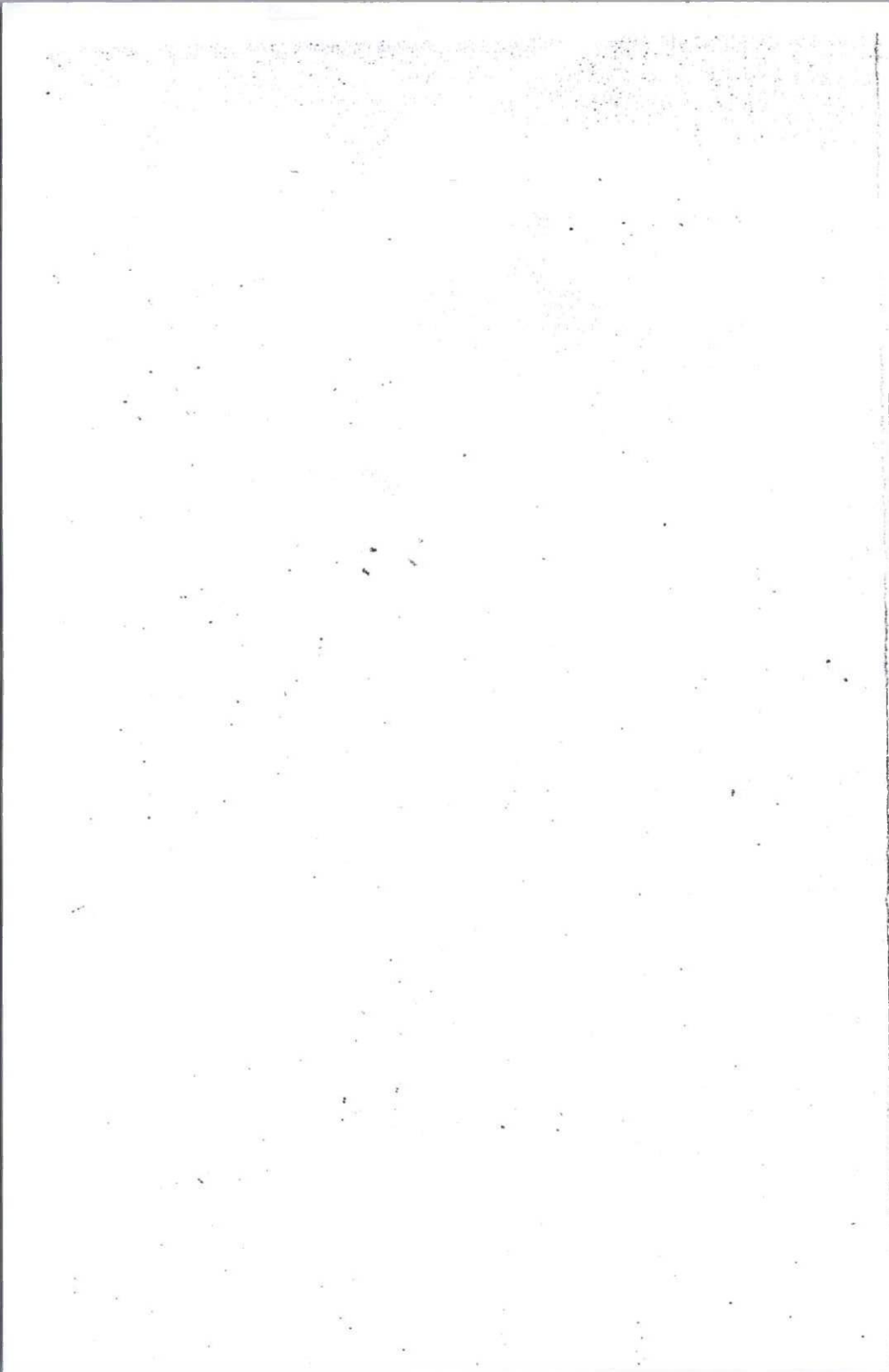
Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Elaborate the Solvent extraction technique with suitable diagram	CO1
(b)	Define the term chromatography and give the classification of Chromatography	
(c)	Explain the crystallization method with suitable diagram	
Q2	(20 marks)	
(a)	Analyze the following terms with suitable example (i) Carbene (ii) Inductive & mesomeric effect	CO2
(b)	Discuss the Orientation effect in aromatic substitution reactions.	
(c)	Elaborate the friedel crafts alkylation reaction of benzene	
Q3	(20 marks)	
(a)	Give the brief introduction of green chemistry and state the twelve Principles of Green Chemistry with suitable example.	CO3
(b)	Give the brief introduction about the nanoparticles and discuss it's type and applications	
(c)	Discuss the top down and bottom up approach of synthesis of nanoparticles and point out the adverse effects associated with the use of harmful chemicals.	
Q4	(20 marks)	
(a)	Explain the principle, advantages and working technique of thin layer chromatography	CO4
(b)	Explain the following terms (i) Molarity, Molality, Normality (ii) Qualitative and Quantitative analysis	
(c)	Elaborate the basic principle and working technique of column chromatography.	
Q5	(20 marks)	
(a)	What do you mean by the term carbohydrate and discuss it's classification with suitable example.	CO5
(b)	Discuss the preparation, physical and chemical properties of glucose.	
(c)	Explain the reactions involve in Kiliani Fischer Synthesis and Ruff degradation.	



END SEMESTER EXAMINATION 2024

Name of the Program: B.Tech. (EE).

Semester: 2nd

Name of the Course: Basic Electrical Engineering

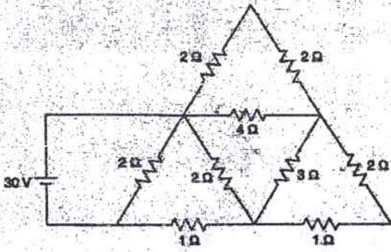
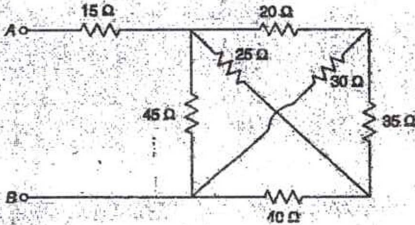
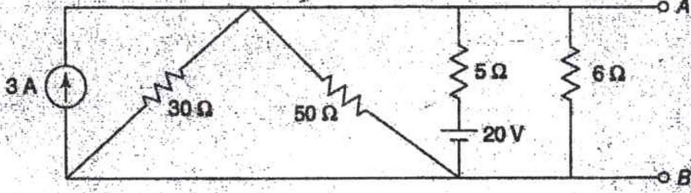
Course Code: TEE201

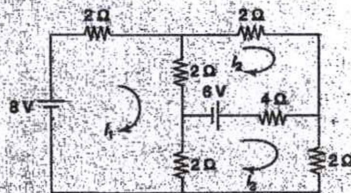
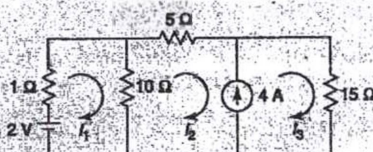
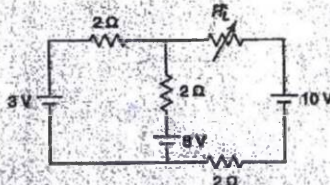
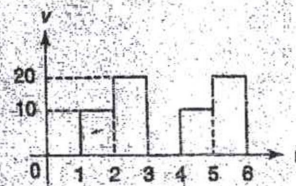
Maximum Marks: 100

Time: 3 hrs

Note

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

1		20 Marks
(a)	Determine the current delivered by the source in the network shown in Fig. 1(a).  Fig. 1(a)	CO1 CO2
(b)	Find an equivalent resistance between A and B in the network shown in Fig. 1(b)  Fig. 1(b)	CO1, CO2,
(c)	Reduce the network shown in Fig. 1(c) into a single source and a single resistor between terminals A and B using source transformation.  Fig. 1(c)	CO1, CO 2 CO 3

2		20 Marks
(a)	Find the current through the 4-ohm resistor in the network of Fig. 2(a) using mesh analysis.	CO2, CO3
	 Fig. 2(a)	
(b)	Find the current through the 10 ohm resistor of the network shown in Fig. 2(b) using superposition theorem.	CO2 CO3
	 Fig. 2(b)	
(c)	Find the value of resistance R_L in Fig. 2(c) for maximum power transfer and calculate maximum power.	CO2, CO3
	 Fig. 2(c)	
3		20 Marks
(a)	An alternating current of 50 c/s frequency has a maximum value of 100 A. (a) Calculate its value $\frac{1}{600}$ second after the instant the current is zero. (b) In how many seconds after the zero value will the current attain the value of 86.6 A?	CO2
(b)	Find the average and rms values of the waveform shown in Fig. 3(b).	CO 2
	 Fig. 3 (b)	
(c)	Two currents i_1 and i_2 are given by the expressions $i_1 = 10 \sin\left(\omega t + \frac{\pi}{4}\right)$ and $i_2 = 8 \sin\left(\omega t - \frac{\pi}{3}\right)$. Find (a) $i_1 + i_2$ and (b) $i_1 - i_2$. Express the answers in the form $i = I_m \sin(\omega t \pm \phi)$.	CO3, CO 2
4		20 Marks
(a)	Define the following terms.	CO 3

	Power factor Impedance RMS value Maximum value Instantaneous value Frequency Time period	
)	An inductive coil draws 10 A current and consumes 1 kW power from a 200 V, 50 Hz ac supply. Determine (a) impedance in rectangular and polar forms, (b) power factor, and (c) reactive and apparent power.	CO 3
)	Define the voltage triangle, power triangle and impedance triangle for single phase series RC circuit.	CO 3
		20 Marks
)	What are the different types of batteries, and their important characteristics? Explain the characteristics of Lithium-ion batteries in detail.	CO4
)	What are the different criterion to select a battery for appropriate application? What are the essential features of switchgear? Also give classification of switchgear based on voltage level.	CO4
)	Explain the need of earthing. Also explain the different types of electrical earthing.	CO5,

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End Semester Examination June 2024

Name of the Course: B.Tech. Biotechnology / B.Sc. Biotechnology

Semester: II

Name of the Paper: Entrepreneurial Biotechnology

Paper Code: TBT 201 / BSBT 207B

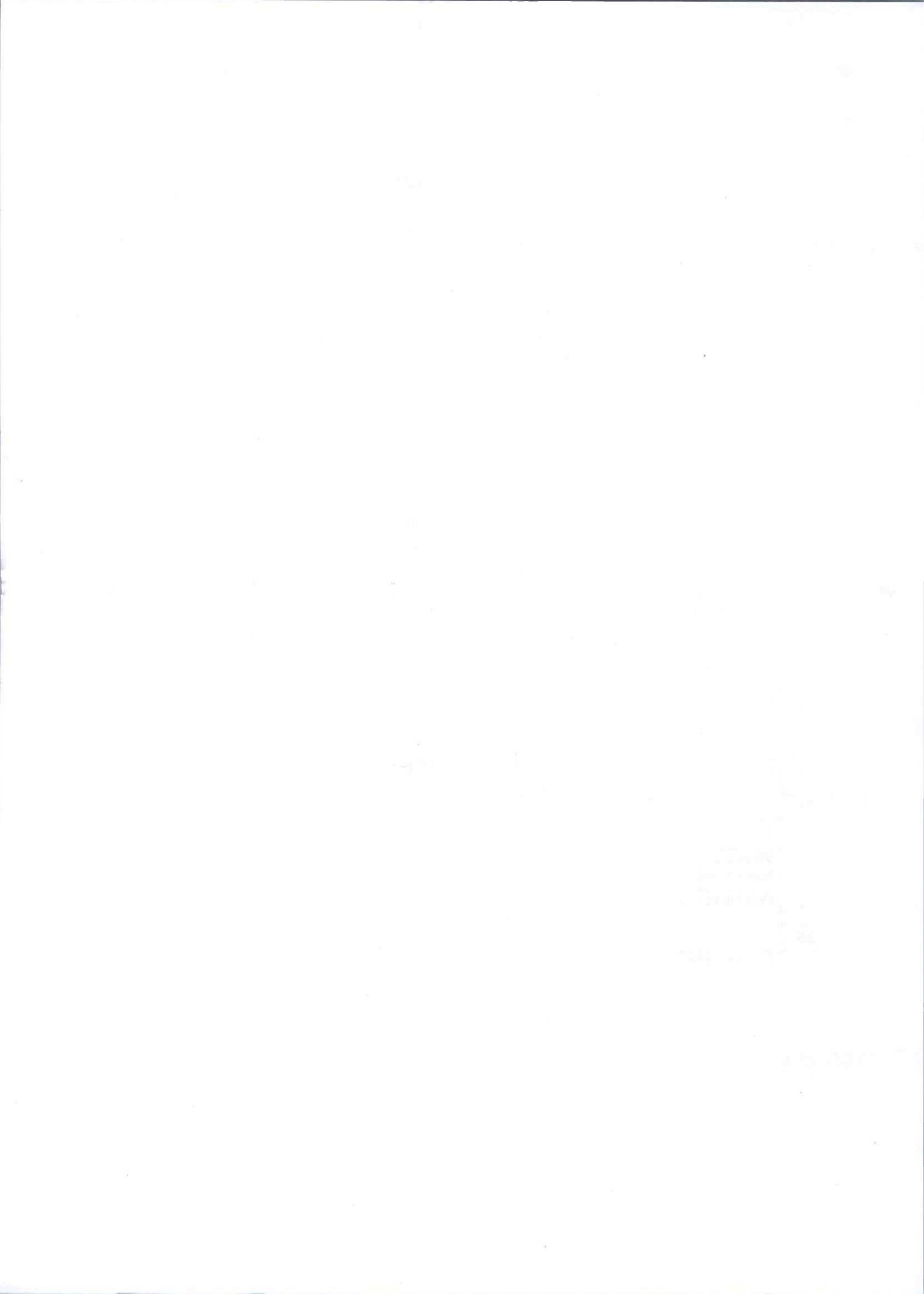
Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1		(20marks)	
(a)	Elaborate on the Cartagena Protocol on Biosafety.		CO1
(b)	What is biosafety? Give a detailed requirement for establishment of a Biosafety Level 4 laboratory.		
(c)	Elaborate on the environmental issues related to bioethics.		
Q2		(20 marks)	
(a)	What is Entrepreneurship? What are the different types of Entrepreneurship?		CO2
(b)	Explain the process and stages in an entrepreneurial startup.		
(c)	What is SWOT analysis? Explain using a biotechnology industry as an example.		
Q3		(20 marks)	
(a)	What is micropropagation? Elaborate on the different stages during the process of micropropagation.		CO3
(b)	What are herbal drugs? Explain how the idea of herbal drug production is a viable bio-business opportunity.		
(c)	What is soil less cultivation of plants? Explain its different types, importance and advantages.		
Q4		(20 marks)	
(a)	What is biocontrol? Discuss the different advantages it has over synthetic pesticides.		CO4
(b)	What is a nutraceutical? Discuss in detail how probiotic based nutraceutical are beneficial to humans with examples.		
(c)	What is vermicompost? Elaborate on its production, use and benefits for growth of plants.		
Q5		(20 marks)	
(a)	Discuss some of the challenges faced by the Biotechnology business in India.		CO5
(b)	Discuss the role of Biotech parks in encouraging bio-business setup		
(c)	Elaborate on the role of the Department of Biotechnology in promoting bio-business startup.		



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End Semester Examination June 2024

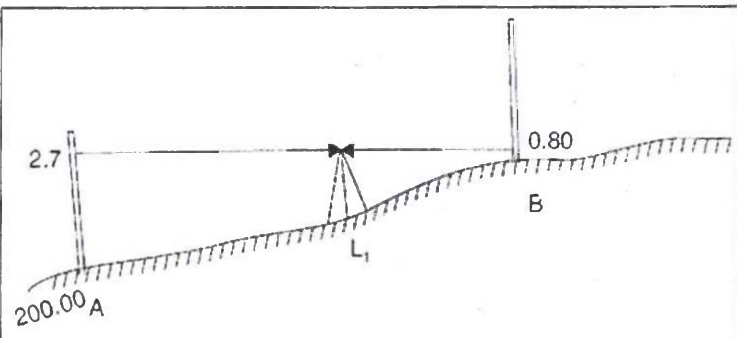
Name of the Course: B. Tech
Name of the Paper: Basic Civil Engineering
Time: 3:00 Hours

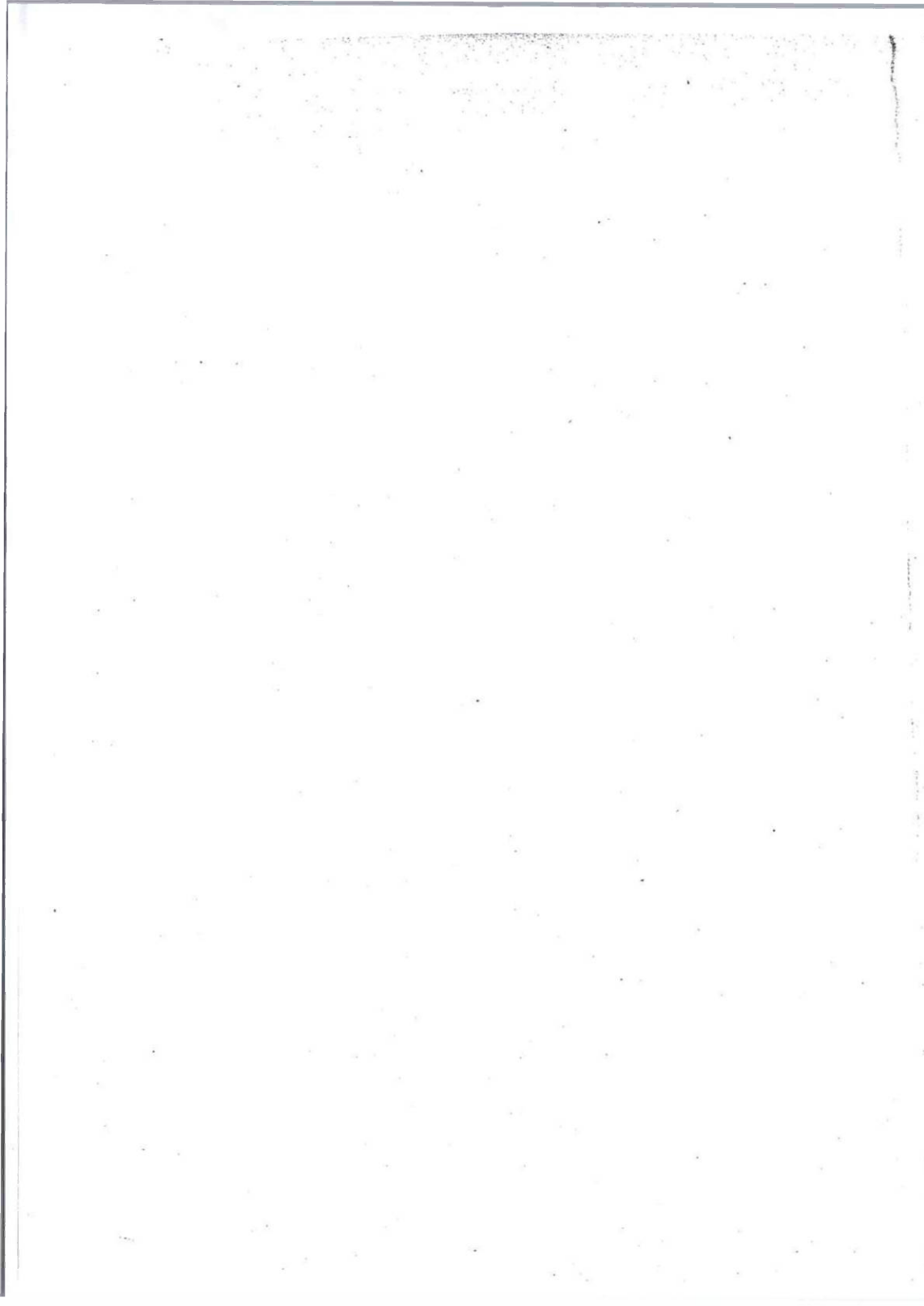
Semester: II
Course Code: TCE 201
Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each question are twenty.
- (iv) Each question carries 10 marks.

Q1		(20marks)				CO1												
(a)	Discuss the following branches of Civil Engineering in terms of their role and responsibilities (i) Geotechnical Engineering (ii) Environmental Engineering (iii) Transportation Engineering (iv) Water Resource Engineering (v) Structural Engineering																	
(b)	Write in detail about any five civil engineering marvels in India?																	
(c)	List down 5 different types of civil engineering structures and their utilization to serve human establishment and society at large.																	
Q2		(20 marks)				CO2												
(a)	In traversing in clockwise direction, the following readings were observed: <table><tr><td>Line</td><td>AB</td><td>BC</td><td>CD</td><td>DE</td><td>EA</td></tr><tr><td>Fore Bearing</td><td>60°30'</td><td>122°0'</td><td>46°0'</td><td>205°30'</td><td>300°0'</td></tr></table> Draw a neat sketch of the traverse. Determine the interior angles of the traverse and apply check.						Line	AB	BC	CD	DE	EA	Fore Bearing	60°30'	122°0'	46°0'	205°30'	300°0'
Line	AB	BC	CD	DE	EA													
Fore Bearing	60°30'	122°0'	46°0'	205°30'	300°0'													
(b)	Differentiate the following (i) Plan and Map (ii) Triangulation and traversing (iii) Chain and tape (iv) Prismatic compass and Surveyor's compass (v) Compass and Theodolite																	

(c)	Find out the R.L. of Point B, with reference to the following figure. 	
Q3	(20 marks)	
(a)	Discuss the following types of cement (i) Ordinary Portland cement (ii) Rapid Hardening Portland cement (iii) High Alumina Cement (iv) Portland Pozzolana Cement (v) Portland Slag Cement	CO3
(b)	Differentiate Timber with Wood. Draw the cross-section of trunk of timber and show the components.	
(c)	Write a short note on Dry manufacturing process of cement. Explain any two field tests for assessing the quality of cement?	
Q4	(20 marks)	
(a)	Briefly discuss the following types of buildings as per NBC (i) Educational Building (ii) Institutional Building (iii) Hazardous Building (iv) Residential Building (v) Assembly Building	CO4
(b)	Discuss the following types of foundations with sketch or line diagram (i) Isolated footings (ii) Well foundation (iii) Combined footings (iv) Strip footings (v) Pile foundation	
(c)	What are the various factors that needs to be kept in mind while planning a building?	
Q5	(20 marks)	
(a)	What are various Traffic Signals? Also, Sketch five regulatory signs, their name and meanings.	
(b)	In the waste water treatment, discuss the (i) Primary treatment (ii) secondary treatment (iii) tertiary treatment.	CO5
(c)	Define Air, Noise and water pollution and enlist the causes of water pollution also explain the various controlled taken to reduce water pollution?	



Roll No.

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2ND END SEMESTER EXAMINATION JUNE - 2024

Name of the Course: B.Tech

Semester: II

Name of the Paper: Programming for problem solving Paper Code: **TCS201**

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Q1 (a) part is compulsory and chose any one from (b) and (c) part.
- (iii) Q2 both (a) and (b) parts are compulsory
- (iv) Answer any two sub questions among (a), (b) & (c) in Q3, Q4 and Q5.
- (v) Total marks in each main question are twenty.
- (vi) Each question carries 10 marks

Q1	(20marks)	
(a)	Create a C program to check whether a given square matrix of order N×N is a lower triangular matrix or not. A lower triangular matrix is the one in which all the element above principal diagonal is zero.	CO1
(b)	Draw a flowchart to pass to string into a function along with start index and end index. Reverse the substring lie between including start index and end index. Print final string in main.	
	Sample Input	Sample output
	Input String: Dehradun	Output string: Deudarhn
	Start index: 2	
	End Index: 6	
(C)	Draw a flowchart to create a structure <i>Person</i> with data member <i>Person_name</i> (char type), <i>Person_age</i> (int type), <i>Person_gender</i> (char type), <i>Person_id</i> (int type) and <i>Person_working_status</i> (string). Read records for 100 Persons and display the detail of persons where: i) persons are female ii) persons are 'non-working' iii) age of persons is between 30-40	
Q2	(20 marks)	
(a)	Write short notes on following by giving suitable example: i) structure v/s array ii) structure within structure iii) self-referential structure iv) bit-field	CO4
(b)	Assuming that the given segments of C and python programs does not prompt any compile time and/or run time errors, predict the output for them, when the following segments of code are executed on a 32-bit machine.	
	(1) #include <stdio.h> #define SIZE 100 int main(void) { char *s1="GEU"; char s2[SIZE]="GEU"; if(s1==s2) printf("EQUAL\n"); else	(2) #include <stdio.h> int main() { FILE *f1,*f2; f1=fopen("hello.txt","a"); f2=fopen("hello.txt","w"); if((f1-f2)) printf("%d\n",A); else
		CO1, CO3, CO4, CO5, CO6

	<pre>printf("UNEQUAL\n"); }</pre>	<pre>printf("%u\n", 'a'); }</pre>	
	(3) (python) <pre>i = 1 while True: if i%64 == 0: break print(i) i = i+i</pre>	(4) (python) <pre>for i in [1, 4, 7, 2, 3, 6, 5][::-1]: print(i)</pre>	
	(5) <pre>#include <stdio.h> typedef struct plantation { unsigned int number_of_trees; unsigned int year; unsigned int area_code; }plantation; int main() { plantation a={100,2024,51}, b={150,2024,50}, *p=&a; printf("%u, %u, %u, %u",sizeof(p), sizeof(a), sizeof(p->area_code), h.year); }</pre>	(6) <pre>#include <stdio.h> void fun(int*); int main() { int i = 10, *p = &i; fun(p++); } void fun(int *p) { printf("%d\n", *p); }</pre>	
	(7) <pre>#include<stdio.h> int main() { char str[] = "GRAPHIC ERA"; char *p=str; *(p+sizeof(str)-5)='\$'; printf("%s\n", str); return 0; }</pre>	(8) <pre>#include <stdio.h> int main() { int a=5,b=6,c=0,*p=&b,*q=&c, *r=&a; *p+=*q-*r; *q=2*(*p) / *r; printf("%d, %d, %d\n", a,b,c); printf("%d, %d, %d\n", *p,*q,*r); }</pre>	
	(9) <pre>#include <stdio.h> typedef struct date { unsigned int dd:5; unsigned int mm:4; unsigned int yy:11; }date; int main() { date d1={31,12,2024}; date d2={17,17,17}; printf("Sizeof date is: %d\n",sizeof(struct date)); printf("d1 is: %d:%d:%d\n",d1.dd,d1.mm,d1.yy); printf("d2 is: %d:%d:%d\n",d2.dd,d2.mm,d2.yy); }</pre>	(10) (python) <pre>def thrive(n): if n % 15 == 0: print("thrive") elif n % 3 != 0 and n % 5 != 0: print("neither") elif n % 3 == 0: print("three") elif n % 5 == 0: print("five") thrive(35) thrive(56) thrive(15) thrive(39)</pre>	
Q3	(20 marks)		C03
(a)	Explain static memory allocation and dynamic memory allocation. What are the advantages of dynamic memory allocation over static memory allocation?		

	Describe various functions to achieve dynamic memory allocation with suitable syntax.		
(b)	Write a program to input n integer elements in an array created by dynamic memory allocation. Check if elements in array are same or not if read from front and back. Print message "Array is same" or "Array is not same" accordingly.		
	Sample Input:	Sample Output:	
	Number of elements in array: 6 Elements in array: 2 3 4 4 3 2	Array is same	
	Number of elements in array: 5 Elements in array: 5 4 3 6 5	Array is not same	
(c)	Write a C code to pass a string into a user define function using call by reference. The task of function is to remove all the characters except alphabets from the string. Print the final string in the calling function.		
	Sample Input:	Sample Output:	
	Input String: program1.c.data@gmail.com	Output String: programcdatagmailcom	
Q4		(20 marks)	
(a)	Describe file. Differentiate binary file v/s text file. What do you mean by sequential file access and random file access? Explain various functions of C to achieve random file access with suitable syntax and explanation.		C05
(b)	Write a C program to write detail of 10 employees into a file named INPUT_DATA.txt. The detail of employee include employee_name, employee_id, employee_basic_salary and employee_gender. Read the file INPUT_DATA.txt and display the detail of all employee along with their Gross_Salary. Gross_Salary = Basic_Salary + DA + HRA DA = 30% of Basic_Salary HRA = 20% of Basic_Salary		
(c)	Write a C program to write n characters into a file named DATA.txt. Read this file and copy content of this file to another file named OUT.txt in reverse order.		
	Sample Input: Data.txt	Sample Output: Out.txt	
	Content of DATA.txt: dehradun	Content of OUT.txt: nudarhed	
Q5		(20 marks)	
(a)	Write short notes on following: i) procedural v/s object oriented ii) compiled language v/s Interpreted language iii) Datatypes in Python iv) Applications of Python		C06
(b)	In a currency system the only available notes are of Rs. 500, Rs. 100, Rs. 50, Rs. 20, Rs. 5 and Rs. 1. Mr. X has to pay amount 'm' to a shopkeeper. Write a code in Python to find the minimum number of notes to make amount 'm'.		
	Sample Input:	Sample Output:	
	Amount to pay: 1283	Minimum number of notes: 11	
(c)	Write a code in Python to generate following pattern: * ** *** **** ***** ***** Till n lines.		